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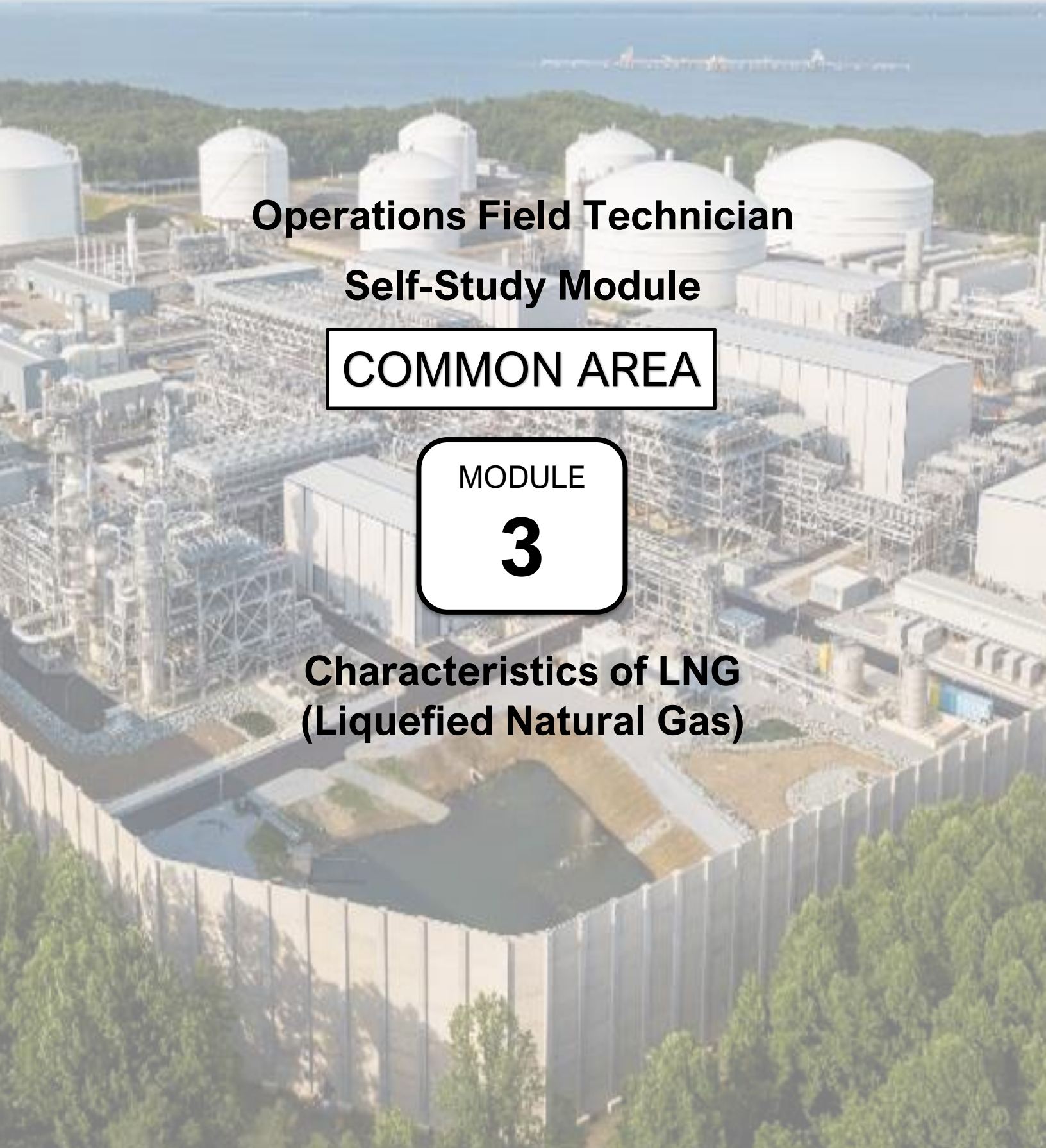
Cove Point LNG Terminal



Operations Field Technician

Self-Study Module

COMMON AREA

MODULE

3

Characteristics of LNG (Liquefied Natural Gas)

Operations Field Technician

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Cove Point LNG Terminal
2100 Cove Point Rd, Lusby, MD 20657

2022

MODULE

3

Characteristics of LNG (Liquefied Natural Gas)

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MODULE 3 OVERVIEW

This self-study module is designed to equip Operations Field Technicians with the knowledge and familiarization of the properties associated with LNG, how to be prepared to recognize process hazards, how LNG can produce cryogenic burns, how to handle cryogenic liquids, and how to wear the proper personal protective equipment (PPE).

The possibility of large spills of liquefied natural gas (LNG) from large, insulated storage tanks or ocean-going LNG tank ships has received increased attention as the demand for natural gas has increased. This attention has resulted because:

- LNG has hazardous properties,
- LNG is stored at extremely low temperatures in large, insulated storage tanks, and
- LNG is transported in large, highly visible ships.

LNG spills are considered serious events because the spill surface provides a large heat source that will rapidly vaporize the LNG. The resultant flammable vapor cloud can travel downwind before diluting to safe concentrations. If the flammable vapor is ignited, a vapor cloud fire followed by a pool fire can occur.

Learning objectives are provided to guide you to the major points in the material and to help you gauge your understanding of the content and your progress. To obtain maximum benefit, each person completing this self-study module should study the material covered and answer the review questions when finished. Each question is designed to address the listed associated learning objectives.

Learning Objectives for this Self-Study Module

1. State the chemical compounds of hydrocarbons.
2. Describe the physical properties of hydrocarbons.
3. Name the three states of matter.
4. Describe the properties of gases.
5. Describe the key definitions of the characteristics of LNG.
6. Discuss the fire hazards of LNG.
7. Describe the types of LNG hazards.

Module Layout

This self-study module provides the basis for training in your specific job responsibilities as they relate to LNG. To safely complete all assigned tasks, you must have an in-depth knowledge of how the process works and operates.

Each lesson in this module builds on the foundation established in the previous lesson. These lessons include a general description including the gas flow path, major equipment, and system operation and control.

Major equipment components are listed in this document with their proper name and Tag Number, i.e., Feed Gas Heater (5E101). Tag numbers are used to standardize identification of those components in drawings, process flow diagrams, P&IDs (Piping and Instrumentation Diagrams) (Piping and Instrumentation Diagrams) and on the actual equipment in the field.

Notes, Cautions, Warnings, and Environmental information may be listed throughout each lesson. This is critical information related to operations, personnel safety, and environmental protection. Read each as it is presented. Notes, Cautions, Warnings, and Environmental Information are described as shown in the legend below.

LEGEND

NOTE

A **NOTE** is used to provide additional information that may be helpful.

CAUTION!

A **CAUTION** is used when the possibility of equipment damage or a serious process upset may occur.

WARNING!

A **WARNING** is used when the possibility of injury or death exists.

Environmental Information

Environmental Information is used to provide specific information or references to Environmental Policy and/or Procedures.

At the conclusion of this self-study module, you will be given a test that encompasses the module information. You will be expected to achieve a minimum passing score.

Glossary of Terms

The glossary provides definitions of major terms used in this module. Familiarity with these terms will assist you in performing your job effectively.

TERM	DEFINITION
CFM	Cubic Feet per Minute
Cryogenic	In Physics, *Cryogenics* is the study of the production and behavior of materials at extremely low temperatures (below –292.00°F).
Emergency	An event that disrupts normal operating conditions and can pose a threat to human life, health, property, or the environment, if not immediately controlled.
Hydrocarbons	Substance made of hydrogen and carbon (among other elements). Common hydrocarbons found in Natural Gas (AKA "Fuel Gas" @ Cove Point LNG): • Methane • Ethane • Propane • Butane • Pentane
LNG	Liquefied Natural Gas: Natural gas composed primarily of methane that is cooled to a temperature of about –260°F at atmospheric pressure where it condenses to a liquid.
LPG	Liquefied Petroleum Gas (LPG), called simply "propane" or "butane," are flammable mixtures of hydrocarbon gases stored in pressurized containers.
MSDS/SDS	Material Safety Data Sheet / Safety Data Sheet
psi	Pounds per Square Inch
psig	Pounds per Square Inch (Gauge)

INTRODUCTION

This self-study module is an introduction to the characteristics of LNG. Understanding the overall function and characteristics of LNG provides you with the information required to work with LNG safely and economically.

In this module, we review the following main topics:

- Hydrocarbons
- States of Matter
- Properties of Gases
- Characteristics of LNG
- Physical Properties of Hydrocarbons
- Hazards of LNG
- Understanding LNG Fire Hazards
- Types of LNG Hazards

SAFETY NOTE

All personnel shall comply with the requirements of the Cove Point LNG *Accident Prevention Manual*. Specific hazards associated with this discussion are:

- Asphyxiation
- Fire
- Cryogenic Liquid (-238°F)

HYDROCARBONS

What are "hydrocarbons"? Any compound made from carbon and hydrogen is called a ***hydrocarbon***.

The carbon and hydrogen atoms are joined by chemical bonds to form hydrocarbon molecules.

Physical Properties of Hydrocarbons

The properties of each type of hydrocarbon depend on the number of carbon and hydrogen atoms in the hydrocarbon molecules, and how they are joined.

Hydrocarbon is the name given to the large family of chemical compounds made up entirely of hydrogen and carbon. Crude oil is a mixture composed almost entirely of hydrocarbons, with a few other compounds like hydrogen sulfide (H_2S) present in some cases.

Learning Objective 1

State the chemical compounds of hydrocarbons.

There are millions of different ways in which carbon and hydrogen atoms can be connected to form hydrocarbon molecules. To make it easier to classify them, these materials are grouped into several large families, e.g., the "paraffin" family."

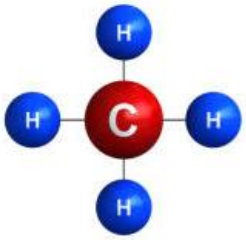
The smallest hydrocarbon molecule is **Methane**, which is the major constituent of natural gas. It has one carbon atom connected to four hydrogen atoms. Its chemical formula is represented as CH_4 and its molecular structure is shown below in **Figure 1**.

Learning Objective 2

Describe the physical properties of hydrocarbons.

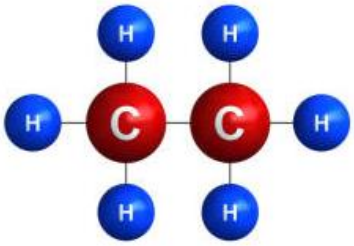
- In any hydrocarbon, the carbon atoms behave as though they each have four arms and the hydrogen atoms behave as though they each have one arm.
- Each arm of the carbon atom must always be occupied, i.e., it must be holding something, either a hydrogen atom, another carbon atom, or an atom of some other element.

- **Methane** — The simplest member of the paraffin family.



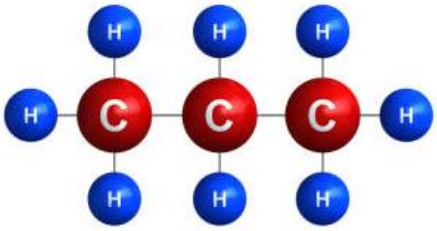
Methane CH_4

- **Ethane** — By adding one carbon atom and two hydrogen atoms, methane is converted to **ethane**.



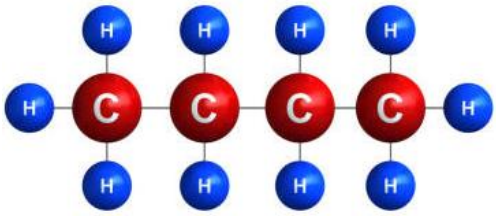
Ethane C_2H_6

- **Propane** — If one more carbon atom and two more hydrogen atoms are added, the result is **propane**, the third member of the paraffin family.



Propane C_3H_8

- **Butane** — The **butane** molecule is composed of four carbon atoms.



Butane C_4H_{10}

Figure 1: Physical Properties of Hydrocarbons

Boiling Point

The normal ***boiling point*** is "the temperature at which a liquid boils at atmospheric pressure." The boiling points of hydrocarbons are related to the number of carbon atoms. The more carbon atoms, the higher the boiling point. The difference in boiling points makes possible the separation of different hydrocarbons (distillation and fractionation).

Boiling Point and Pressure

The boiling point of a compound is affected by the pressure at which the compound is held, e.g., propane boils at –43.8°F @ atmospheric pressure; however, if contained in a system at about 174 psi, the boiling point rises to +95°F.

Vapor Pressure

Any liquid is made up of many molecules, all of which are in motion. Some of these molecules have enough energy to break through the liquid's surface into the gaseous state. Consider an open container partially filled with liquid.

- The molecules that break free of the liquid can escape completely and the liquid will eventually evaporate.
- However, in a closed container, the increasing number of gaseous molecules creates an increase in pressure. It also increases the tendency for the molecules to re-enter the liquid.

Equilibrium is reached when the number of molecules leaving the liquid state equals the number re-entering the liquid. The pressure developed at this equilibrium is known as the *vapor pressure* of the liquid.

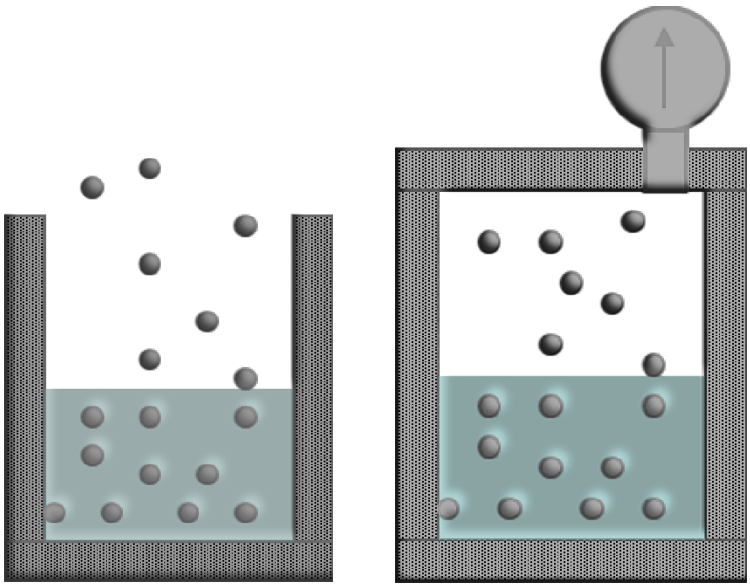


Figure 2: Open/Closed Container

Vapor Pressure and Temperature

The relationship between vapor pressure and temperature can be understood in terms of molecules in the liquid escaping the attractive forces of the liquid to enter the gas phase.

Boiling Point and Vapor Pressure

When a liquid is heated in an open vessel, the ***vapor pressure*** will increase until it reaches the atmospheric pressure surrounding the liquid. When this happens, it is accompanied by the physical phenomena of boiling, as vapor bubbles form in the liquid and rise to the surface.

The ***boiling point*** of a liquid is "the temperature at which its vapor pressure is equal to the surrounding pressure."

Specific Gravity

Specific gravity of a liquid is "a ratio of the weight of a volume of the liquid to the weight of an equal volume of water at a designated temperature."

Specific gravity of a gas is "a ratio of the weight of a volume of the gas to the weight of an equal volume of air at a designated temperature and pressure."

Table 1: Specific Gravity

Parameter	Methane C1 (CH_4)	Ethane C2 (C_2H_6)	Propane C3 (C_3H_8)	Iso-Butane iC4 (C_4H_{10})	N-Butane nC4 (C_4H_{10})
Molecular Weight	16	30	44	58	58
Liquid Specific Gravity	0.3	0.36	0.51	0.56	0.58
Vapor Specific Gravity	0.55	1.04	1.52	2.01	2.01

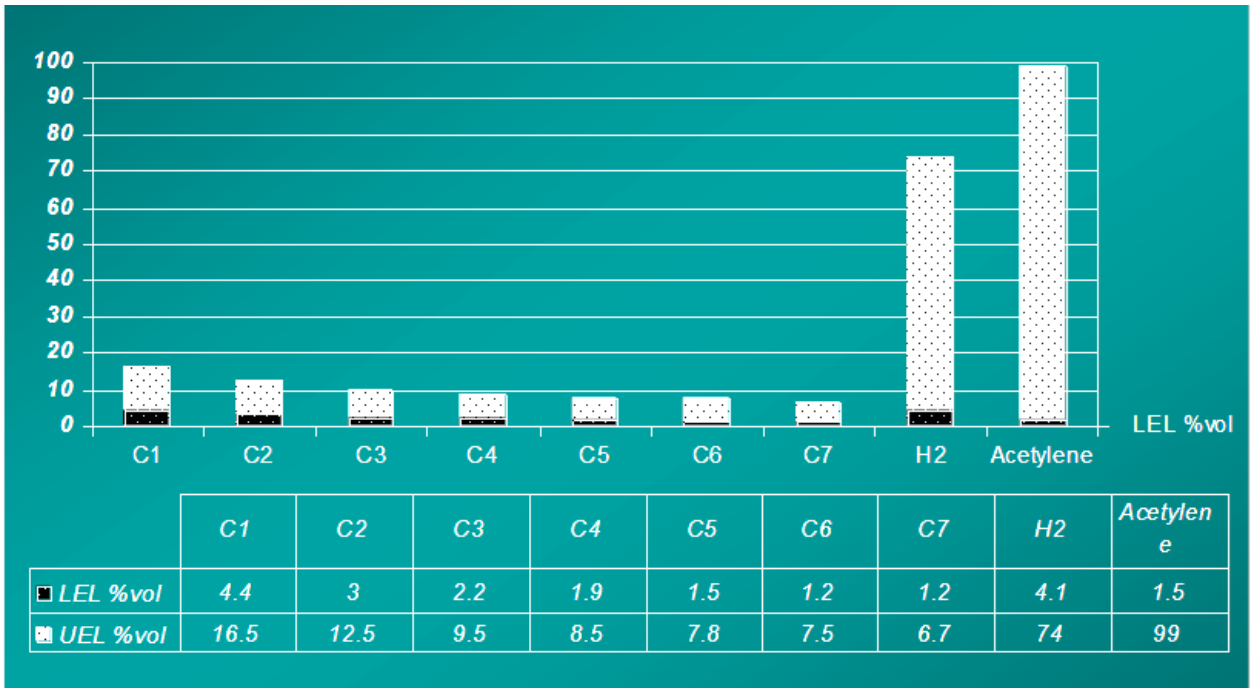


Figure 3: Flammability Limits of Hydrocarbons

Table 2: Physical Properties of Hydrocarbons

Parameter	C1 Methane	C2 Ethane	C3 Propane
Chemical Formula	**CH_4**	C_2H_6	$C3H_8$
Molecular Weight	16	30	44
Boiling Point	**–259.6°F**	–128.2°F	–43.6°F

All the above hydrocarbons have a general formula of CnH2n+2 and belong to the classification of hydrocarbons called “paraffins.”

States of Matter

All matter exists in one of three states: solid, liquid, or a gas (refer to **Figure 4**).

Learning Objective 3

Name the three states of matter.

1. In a **solid**, molecules are close together and are slow-moving.
2. In a **liquid**, molecules are further apart and move faster than in a solid.
3. In a **gas**, molecules are far apart and move very rapidly.

The rate of molecule movement determines whether matter is a solid, a liquid, or a gas.

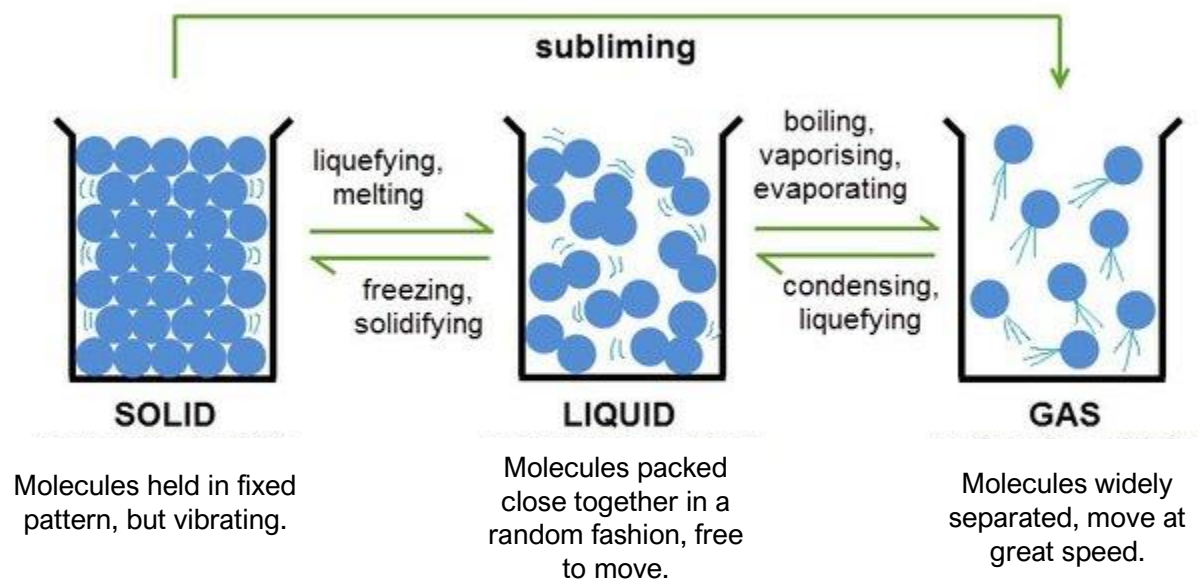


Figure 4: Three States of Matter

- **Solids** do not need a container as they have a definite shape and volume of their own.
- **Liquids** have a definite volume, but no definite shape. Liquids take the shape of whatever container they are put in.
- **Gases** have no definite shape or volume and expand to fill a container, no matter what its size or shape. A gas not enclosed in a container keeps on expanding until its molecules become so far apart, it is impossible to detect the gas.

Properties of Gases

Learning Objective 4

Describe the properties of gases.

Gas is a state of matter that has neither definite shape nor volume but tends to expand indefinitely.

The properties of a gas are Volume, Pressure, and Temperature.

The property changes of a gas are explained by the *Ideal Gas Laws*, which are briefly reviewed in this section. These are called the *Ideal Gas Laws* because no actual gas obeys it exactly, although all obey it over a wide range of conditions.)

Boyle's Law

Boyle's Law states that, for any *isothermal* (same temperature) process, **the volume of a fixed mass of gas is inversely proportional to its absolute pressure"** (refer to **Figure 5**).

$$P \ \alpha \ \frac{1}{V} \qquad \textbf{OR} \qquad PV = \text{a constant (F)}$$

Remember that all units should be 'absolute' and that:

Absolute pressure = Gauge pressure + Atmospheric pressure

Simply stated, *Boyle's Law* states that: **At constant temperature, if the volume of a gas is reduced by half, its pressure is doubled.**

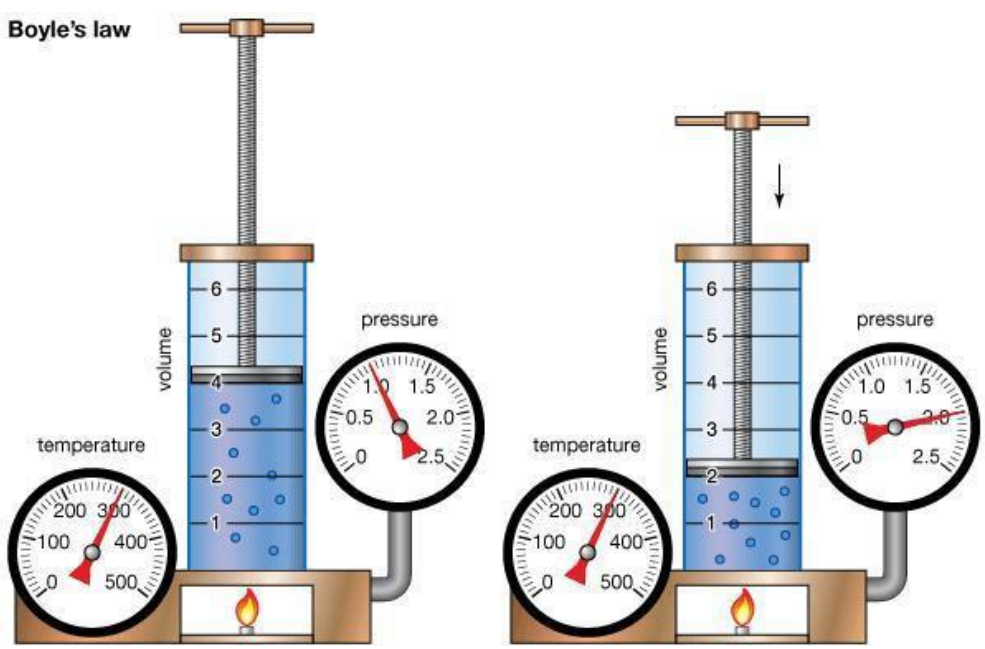


Figure 5: Boyle's Law

Charles' Law

Charles Law states that at a **Constant Pressure**, if the absolute temperature is doubled, then its volume will also be doubled.

For any *isobaric* (same pressure) process, the **volume of a given mass of gas is directly proportional to its absolute temperature** (refer to **Figure 6**).

This can be written as: $V \ \alpha \ T$ **OR** $\frac{V}{T} = \text{a constant}$

Simply put, this means that, at the same pressure, **if the absolute temperature of a gas is increased by double, its volume will also be doubled.**

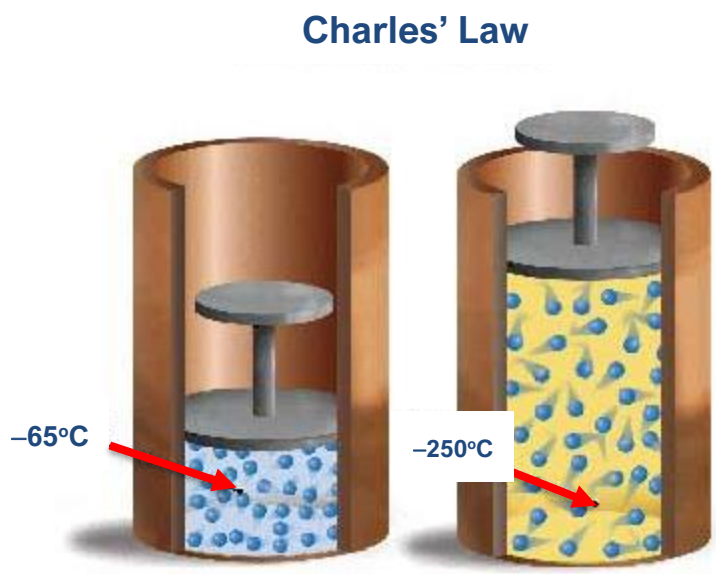


Figure 6: Charles' Law

Combined Gas Law

Boyle's Law and *Charles' Law* refer to "isothermal" and "isobaric" changes, respectively. However, during many gas processes, the gas pressure and temperature change simultaneously. Therefore, it is desirable to use a formula that relates to the three variables of pressure, temperature, and volume.

The relationship between these is referred to as the *Combined Gas Law*, which is written:

$$\frac{P_1V_1}{T_1} = \frac{P_2V_2}{T_2}$$

In hydrocarbon processing, all gases are contained in pipelines and vessels. This makes it possible to calculate the volume of gas in a vessel or system. To remove the effects of temperature and pressure on the volume of the gas, all calculations are made assuming a standard temperature of 60°F and a pressure of 14.7 psia (air pressure at sea level).

The following properties of gases are described below.

- **Density** — *The density* of a gas is mass per unit volume measured in pounds per cubic foot at Standard Temperature and Pressure (STP) (60°F) and 14.7 psi. Even if the volume of a gas is measured under different conditions, it is not difficult to calculate its volume at STP, using the gas laws.
- **Compressibility** — This is the ability to be compressed in volume. All gases can be compressed. Compressing a gas reduces its volume by squeezing the gas molecules together, thereby making it hotter.
- **Flammability** — This is the ability to burn in air. Hydrocarbon gases are flammable.
- **Inertness** — This is the property whereby a gas does not react chemically with other matter. Nitrogen is an inert gas used to remove air or hydrocarbons from equipment. An inert gas is not flammable under normal conditions.

Liquefied Natural Gas (LNG) Properties

Learning Objective 5

Describe the key definitions of the characteristics of LNG.

Natural gas produced from the wellhead consists of methane, ethane, propane, and heavier hydrocarbons, plus small quantities of nitrogen, helium, carbon dioxide, sulfur compounds, and water. ***LNG*** is *liquefied natural gas*.

The LNG Liquefaction process first requires pre-treatment of the natural gas stream to remove impurities such as mercury, water, carbon dioxide, hydrogen sulfide, and other sulfur compounds. By removing these impurities, solids cannot be formed while the gas is refrigerated. The final product will then meet the quality specifications of the LNG end users.

Natural gas is liquefied at high pressure and warmer temperature than -256°F. Then, its pressure is reduced to be stored slightly above atmospheric pressure and below -256°F. LNG takes up only 1/600th of the volume required for a comparable amount of natural gas at room temperature and normal atmospheric pressure. Because it is not stored under pressure, the common misperception of LNG as a pressurized substance has led to an erroneous understanding of its danger.

LNG is a clear, non-corrosive, non-toxic, near odorless, cryogenic liquid at normal atmospheric pressure. Although methane is non-toxic, as with any gaseous material besides air and oxygen, natural gas that is vaporized from LNG can cause asphyxiation due to lack of oxygen if a concentration of gas develops in an unventilated, confined area.

LNG FIRE HAZARDS

In this section, we discuss the following main topics:

- Understanding LNG Fire Hazards
- Types of LNG Hazards

Understanding LNG Fire Hazards

Vapors released from LNG as it returns to a gas phase, if not properly and safely managed, can become flammable, but explosive only under certain conditions. Safety and security measures contained in the engineering design and technologies, and in the operating procedures of LNG facilities reduce these potential dangers.

Learning Objective 6

Discuss the fire hazards of LNG.

Flammability Range

The ***flammability range*** is "the range between the minimum and maximum concentrations of vapor (percent by volume) in which air and LNG vapors form a flammable mixture that can be ignited and burn."

Figure 7 indicates that the Upper Flammability Limit (UFL) and Lower Flammability Limit (LFL) of Methane, the dominant component of LNG vapor, are 5% and 15% by volume, respectively.

- When fuel concentration exceeds its UFL, it cannot burn because too little oxygen is present.

  This situation exists, for example, in a closed, secure storage tank where the vapor concentration is near 100% methane.

- When fuel concentration is below the LFL, it cannot burn because too little methane is present. An example is leakage of small quantities of LNG in a well-ventilated area.

  In this situation, the LNG vapor will rapidly mix with air and dissipate to less than 5% concentration.

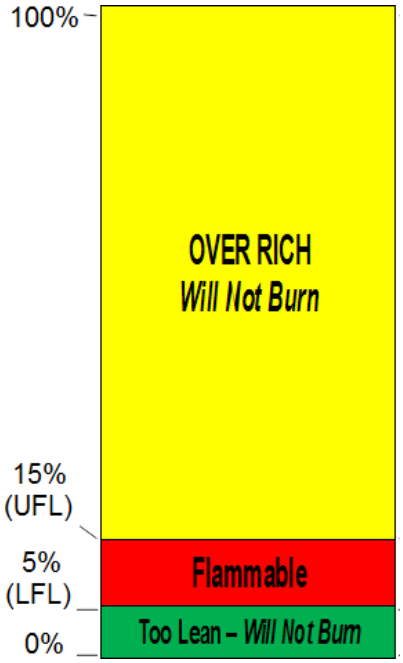


Figure 7: Flammable Range for Methane

Flammability Comparison

A comparison to the properties of LNG to those of other liquid fuels, as shown in **Table 3**, also indicates that the lower flammability limit of LNG is generally higher than other fuels. That is, more LNG vapors would be needed (in a given area) to ignite as compared to LPG or gasoline.

Table 3: Comparison of LNG Properties to Other Liquid Fuels

PROPERTIES	LNG	LIQUEFIED PETROLEUM GAS (LPG)	GASOLINE	FUEL OIL
Toxic	No	No	Yes	Yes
Carcinogenic	No	No	Yes	Yes
Flammable Vapor	Yes	Yes	Yes	Yes
Forms Vapor Cloud	Yes	Yes	Yes	Yes
Asphyxiant	Yes, but in a vapor cloud	Yes, but in a vapor cloud	Yes	Yes
Extreme Cold Temperature	Yes	Yes, if refrigerated	No	No
Other Health Hazards	None	None	Eye irritant, narcosis (unconsciousness), nausea, others	Eye irritant, narcosis (unconsciousness), nausea, others
Flash Point (°F)	-306	-156	-50	140
Boiling Point (°F)	-256	-44	90	400
Flammability Range in Air (%)	5 – 15	2.1 – 9.5	1.3 – 6	N/A
Stored Pressure	Atmospheric	Pressurized (atmospheric if refrigerated)	Atmospheric	Atmospheric
Behavior if Spilled	Evaporates, forming visible "clouds." Portions of cloud could be flammable or explosive under certain conditions	Evaporates, forming visible "clouds." Portions of cloud could be flammable or explosive under certain conditions	Evaporates, forms flammable pool: Environmental cleanup required.	Evaporates, forms combustible pool: Environmental cleanup required.

Ignition and Auto Ignition

An often-unexpected hazard is **ignition** from flames or sparks. Consequently, LNG facilities are designed and operated using standards and procedures to eliminate this hazard, and are equipped with extensive fire detection and protection systems should flames or sparks occur.

The **auto ignition** temperature is the lowest temperature at which a flammable gas vapor will ignite spontaneously, without a source of ignition, after several minutes of exposure to sources of heat.

- Temperatures higher than the auto ignition temperature will cause ignition after a shorter exposure time.
- With exceedingly hot temperatures, and within the flammability range, ignition can be virtually instantaneous.

For methane vapors derived from LNG, with a fuel-air mixture of 10% methane in air (about the middle of the **5–15%** flammability limit) and atmospheric pressure, the auto ignition temperature is above 1,000°F. This extremely hot temperature requires a strong source of thermal radiation, heat, or hot surface.

If LNG is spilled on the ground or on water and the resulting flammable gas vapor does not encounter an ignition source (a flame or spark or a source of heat >1,000°F), the vapor will dissipate into the atmosphere, and no fire will take place.

When compared to other fuels, LNG vapor (methane) requires the highest temperature for auto ignition as shown in the following table:

Fuel	Auto-Ignition Temperature (°F)
LNG (primarily methane)	1,004
LPG	850 – 950
Ethanol	780 - 800
Methanol (Methane)	**867**
Gasoline	495
Diesel Fuel	Approx. 450

Potential Hazards

Potential hazards resulting from intentional or accidental spilling of large quantities of LNG include thermal radiation from **vapor cloud fires** (also referred to as *flash fires*) and **pool fires**. There is general agreement among LNG experts regarding the following aspects of potential LNG fire and explosion hazards:

Vapors from large, unignited spills of LNG cannot travel far into developed areas without finding an ignition source, igniting, and burning back to the source.

Vapor Cloud Fire

Once delayed ignition of the vapor cloud occurs, and provided that the cloud is unconfined and rich in methane, the LNG vapor will burn in the form of a **vapor cloud fire**.

NOTE

A ***vapor cloud fire***, also called "flash fire," refers to the combustion of a vapor cloud without generating any significant overpressure.

A vapor cloud traversing over commercial and/or residential terrain will almost certainly encounter an ignition source early in its downwind drift, and the resulting vapor cloud fire will burn back to the source.

The vapor cloud fire will burn back to the source and cause a pool fire at the source if the release is a continuous release, and the release duration is longer than the time it takes the cloud to find an ignition source.

If the vapor cloud is confined and/or the vapors contain large amounts of heavier hydrocarbons (C2+, ethane, propane, butane, etc.), then the flame can accelerate and result in an explosion. The magnitude of the explosion and explosion damage depends on several factors including the amount of vapors above the lower flammable limit, the presence of obstacles and degree of confinement, the composition of the vapor cloud, and the strength of the ignition source.

Pool Fire

If immediate ignition occurs, a **pool fire** will result. The extent of the pool spreading (diameter), and flame height will depend on several factors including the flow rate of LNG, the spill surface type (water or land), the spill surface geometry, spill surface roughness, release composition, release temperature, ambient wind speed, ambient temperature, ambient relative humidity, and containment pit or bounded area.

NOTE

A ***pool fire*** is a type of diffusion flame where a layer of volatile liquid fuel is evaporating and burning.

If the liquid pool is unconfined and the inventory of LNG is large, the fire will continue to burn until all the fuel is exhausted by the pool fire. It is not practical or even possible to extinguish large LNG pool fires resulting from large spills of LNG unless the flow of LNG feeding the pool can be stopped.

Figure 8: LNG Pool Fire

Calculating Who is at Risk

The maximum vapor cloud fire hazard area from large LNG spills is typically estimated by calculating a downwind dispersion distance to the lower flammable limit (LFL) and a crosswind dispersion distance to 1/2 LFL at low wind speed and stable atmospheric conditions. This maximum fire hazard zone is very unlikely to be experienced in any situation where the cloud drifts overpopulated areas. As indicated above, the cloud will soon encounter an ignition source and burn back to the source well before the maximum hazard area is reached.

Only the outdoor population present within the flammable boundaries of the vapor cloud is assumed to be injured due to (a) short exposure to very high thermal radiation fluxes from the vapor cloud fire, (b) direct flame contact, (c) secondary fires of clothing, and (d) inhalation of hot combustion products.

- It is assumed that people inside buildings at the time of the flash fire will not be injured.
- It is also assumed that people inside buildings ignited by flash fire or pool fire will escape from the burning structure without direct thermal impact injuries. This is because the flash fire will ignite buildings from the outside and it will take some time for the fires to penetrate inside.

Fire Hazards Summary

Questions about LNG safety often demonstrate how LNG is often confused with other fuels and materials. All these fuels can be used safely if proper safety, security, and environmental protections are in place. We fill our cars and trucks with gasoline, use LPG (propane) in our backyard grills, and use methane to heat our homes, hundreds of millions of times each day, and serious safety incidents are rare.

In summary, LNG is an extremely cold, non-toxic, non-corrosive substance that is transferred and stored at atmospheric pressure. It is refrigerated, rather than pressurized, which enables LNG to be an effective, economical method of transporting large volumes of natural gas over long distances. LNG poses little danger if it is contained in storage tanks, piping, and equipment designed for use in LNG cryogenic conditions. However, vapors resulting from LNG because of an uncontrolled release can be hazardous, within the key properties of LNG and LNG vapors – flammability range and in contact with a source of ignition – as described above.

Types of LNG Hazards

Learning Objective 7

Describe the types of LNG hazards.

The potential hazards of most concern to LNG facility personnel and the surrounding community flow from the basic properties of natural gas.

Primary containment (tanks, piping, etc.), secondary containment (dikes, walls, spillways, etc.), safeguard systems (ESD, fire and gas detection, etc.), and separation distance provide multiple layers of protection. These measures provide protection against hazards associated with LNG.

Explosion

An **explosion** happens when a substance rapidly changes its chemical state, i.e., is ignited or is uncontrollably released from a pressurized state. For an uncontrolled release to happen, there must be a structural failure (ruptured pipe, tank, pump, etc.). LNG tanks store the liquid at an extremely low temperature, about -256°F, so no pressure is required to maintain its liquid state. Since LNG is stored at atmospheric pressure, a crack or puncture in the container will not create an immediate explosion.

Vapor Clouds

As LNG leaves a temperature-controlled container, it begins to warm up, returning the liquid to a gaseous state. Initially, the gas is colder and heavier than the surrounding air. It creates a fog – a **vapor cloud** – above the released liquid. As the gas warms, it mixes with the surrounding air and begins to disperse. The vapor cloud will only ignite if it encounters an ignition source while concentrated within its flammability range.

Cryogenic Burns

The effects of cryogenic liquid upon the skin are very rapid and can cause frostbite and freezing. The injury received can be severe, can occur in seconds, and has the same impact as a burn injury, 1st, 2nd, or 3rd degree. This must be kept in mind when responding to LNG or Liquid Nitrogen (LN2) spills.

PPE

The adverse effect of a cryogenic liquid upon the skin can be minimized by wearing the proper personal protective equipment (PPE). These items include, but are not limited to:

- Rain slickers
- Face shields and goggles
- Hard hat
- Rubber coated gloves
- Rubber boots

Severity Identification

The following are general indications and identification of the severity of a cryogenic burn:

- **1st degree burns** can cause numbing of the skin, which may turn white in color. Skin may feel stiff to the touch, but underlying tissue is still soft.
- **2nd degree burns** can cause skin to turn white or blue in color and feel hard and frozen. The underlying tissue is still undamaged; blistering is likely.
- **3rd degree burns** can cause skin to be white, blotchy, and/or blue in color. The underlying tissue is hard and cold to the touch.

Emergency Action Steps

The following steps should be taken immediately to treat cryogenic burns:

1. Immediately flush the affected areas with clean, cool water.
2. Wrap the affected areas with loose, clean bandages.
3. **DO NOT** rub the affected areas.
4. **Seek immediate medical assistance**.

Handling Cryogenic Liquids

Most cryogenic liquids are odorless, colorless, and tasteless when vaporized. When cryogenic liquids are exposed to the atmosphere, the cold boil-off gases condense the moisture in the air, creating a highly visible fog.

ALWAYS handle these liquids carefully. Because of their extremely low temperatures, they can produce cryogenic burns and frostbite. When spilled on a surface, they tend to cover it completely and, therefore, cool a large area. The vapors from these liquids are also extremely cold and can produce burns. Exposure that may be too brief to affect the skin of the face or hands may damage delicate tissues, such as the eyes.

Boiling and splashing always occurs when charging or filling a warm container with cryogenic liquid or when inserting objects into these liquids. Perform these tasks slowly to minimize boiling and splashing. Use tongs to withdraw objects immersed in a cryogenic liquid.

NEVER touch uninsulated pipes or vessels containing cryogenic liquids. Flesh will stick to extremely cold materials. Even non-metallic materials are dangerous to touch at low temperatures. In addition to the hazards of frostbite or flesh sticking to cold materials, objects that are soft and pliable at room temperature, such as rubber or plastic, become hard and brittle, and break easily at these extremely low temperatures.

Rollover

When LNG supplies of multiple densities (various places of origin, age, etc.) are loaded into a tank one at a time, they do not initially mix. Instead, they will layer themselves in unstable strata within the tank. As the lower LNG layer is heated by normal heat leak, it changes density until it finally becomes lighter than the upper layer. At that point, a liquid ***rollover*** will occur, which can cause a sudden increase in vapor generation within the tank. It is possible to experience a rollover so violent as to generate more vapor than the tank's pressure control devices (PSV's, vents, boil-off gas compression) can handle, which can cause cracks or other structural failures in the tank.

To help prevent the layering, or stratification, the operators unloading an LNG vessel, or simply consolidating inventory, monitor the density of both the product being transferred and the product in the receiving tank, and decide whether to fill the receiving tank from the bottom or the top fill connections.

Rapid Phase Transition

When released on water, LNG floats and vaporizes. If large volumes of LNG are released on water, it may vaporize too quickly, causing a ***rapid phase transition*** (RPT) or physical explosion.

The same effect can happen when large amounts of water are introduced to an LNG pool. Water temperature and the presence of substances other than methane also affect the likelihood of an RPT. An RPT can only occur if there is mixing between the LNG and water. A common example of an RPT is the rapid flash to steam when water is added to a pan of hot oil. **Figure 9** is a photo sequence of a deliberate LNG RPT test.

Figure 9: LNG Rapid Phase Transition Test

MODULE SUMMARY

What You Have Learned...

In this module, you learned about the characteristics of LNG. You also learned the effects of exposure to and potential for damage from LNG leaks.

In this module, we reviewed the following main topics:

- Hydrocarbons
- States of Matter
- Properties of Gases
- Characteristics of LNG
- Physical Properties of Hydrocarbons
- Hazards of LNG
- Understanding LNG Fire Hazards
- Types of LNG Hazards

The next module reviews the Electrical Distribution System. ■

Notes